MTH129 Introduction to AI for Mathematical Sciences (MS)

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AI Ethics

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- AI is already happening today and it is pervasive, often invisibly embedded in our day-to-day tools.
- The Need to Discuss Ethical and Societal Problems
  - AI can have many benefits. It can be used to improve public and commercial services. For example, image recognition is good news for medicine: it can help with the diagnosing of diseases such as cancer.
  - But such every-day applications of artificial intelligence also show how the new technologies raise ethical concerns.
  - Should self- driving cars have built- in ethical constraints, and if so, what kind of constraints, and how should they be determined?
  - For example, if a self- driving car gets into a situation where it must choose between driving into a child or into a wall to save the child's life but potentially killing its passenger, what should it choose?
- **AI ethics** is about technological change and its impact on individual lives, but also about transformations in society and in the economy.

ALL ABOUT THE HUMAN

- Is General AI Possible? Are There Fundamental Differences between Humans and Machines?
- There is a history of criticism and skepticism about the possibility of human-like AI.
- Many philosophers think that there are crucial differences between how humans and computers think.
- For example, people are meaning-making, conscious, embodied, and living beings whose nature, mind, and knowledge cannot be explained away by comparisons to machines.
- The discussion about AI is about deep disagreements about the nature of the human, human intelligence, mind, understanding, consciousness, creativity, meaning, human knowledge, science, and so on.

JUST MACHINES?

- Questioning the Moral Status of AI: Moral Agency and Moral Patiency
- Moral Agency: what the AI is capable of doing morally speaking and, if so, whether it can be a full moral agent.
- Moral Patiency: how we should treat an AI.
- Is an AI "just a machine"? Should we treat it differently than, say, a toaster or a washing machine?
- This question is not about the ethics by or in AI but about our ethics toward AI.
- Here the AI is object of ethical concern, rather than a potential ethical agent itself.

Moral Agency

- If an AI were to be more intelligent than is possible today, we can suppose that it could develop moral reasoning and that it could learn how humans make decisions about ethical problems.
- But would this suffice for full moral agency, that is, for human-like moral agency?
- Machines can never be moral agents at all: Machines do not have the required capacities for moral agency such as mental states, emotions, or free will.
- Machines can be full moral agents in the same way that humans are: it is possible and desirable to give machines a human kind of morality (Anderson and Anderson 2011)
- If general AI is possible at all, then we don't want a kind of "psychopath AI" that is perfectly rational but insensitive to human concerns because it lacks emotions (Coeckelbergh 2010)
- We have to give AIs some kind of morality, but not full morality

Moral Patiency

- Imagine that we have a superintelligent AI. Is it morally cceptable to switch it off, to “kill” it?
- Humans seem to require very little of artificial agents in order to project personhood or humanness onto them and to empathize with them.
- If these agents now become AI, which potentially make them more human-like (or animal- like), this seems to make the question regarding moral patiency only more urgent.
- For example, how should we respond to people who empathize with an AI?
- Some argue that “mistreating” an AI is wrong not because any harm is done to the AI, but because our moral character is damaged if we do so.
- Regardless of what the actual moral status of AIs “is,” as if this could be defined entirely independent from human subjectivity, it is worth critically examining our own moral attitude and the project of abstract moral reasoning itself.

Toward More Practical Ethical Issues

- AI is about the less visible, backstage but pervasive, powerful, and increasingly smarter technologies that already shape our lives today.
- AI ethics is about the ethical challenges posed by current and near-future AI and its impact on our societies and vulnerable democracies.
- AI ethics is about the lives of people and it is about policy.
- It is about the need for us, as persons and as societies, to deal with the ethical issues.

Privacy and Data Protection

- AI, and in particular machine learning applications working with big data, often involves the collection and use of personal information.
- AI can also be used for surveillance, on the street but also in the workplace and— through smart-phones and social media— everywhere.
- An ethical use of AI requires that:
  - data are collected, processed, and shared in a way that respects the privacy of individuals and their right to know what happens to their data,
  - to access their data, to object to the collection or processing of their data,
  - and to know that their data are being collected and processed and (if applicable) that they are then subject to a decision made by an AI.
- Data protection issues also arise in research ethics, for example, the ethics of collecting data for social science research.

Manipulation, Exploitation, and Vulnerable Users

- As social media users, we risk becoming the unpaid, exploited workforce that produces data for the AI that then analyzes our data— and in the end for the companies that use the data, which usually also include third parties.
- AI may lead to new forms of manipulation, surveillance, and totalitarianism, not necessarily in the form of authoritarian politics but in a more hidden and highly effective way:
- *by changing the economy in a way that turns us all into smartphone cattle milked for our data.*
- Theories of privacy and exploitation often assume that the user is an autonomous and relatively young and healthy adult human being with full mental capacities.
- The real world is one populated with children, elderly people, people who do not have “normal” or “full” mental capabilities, and so on.
- These vulnerable users are **more at risk**.

Fake News, the Danger of Totalitarianism, and the Impact on Personal Relationships

- AI may also be used to produce hate speech and false information, or to create bots that appear to be people but in fact are AI software.
- False information and manipulation existed before AI.
- Film, for example, has always created illusions, and newspapers have spread propaganda.
- But with AI, combined with the possibilities and environment of the internet and digital social media, the problem seems to increase in intensity.
- With AI, the technology is much better in creating the illusion of companionship and that this increases the risk of loneliness or deteriorating personal relationships.

Safety and Security

- AIs, especially when embedded in hardware systems that operate in the physical world, also need to be **safe**.
- Should a robot always avoid bumping into humans or is it OK if it sometimes obstructs a person in order to reach its goal?
- In a networked world, every electronic device or software can be hacked, invaded, and manipulated by people with malicious intentions.
- When equipped with AI, our devices and software can do more, and when they gain more agency and this has real-world physical consequences, the **security** problem becomes much larger.
- Technological vulnerabilities ultimately transform our existence as humans.
- Increased agency of AI, especially when it replaces human agency, also raises another ever more urgent ethical problem: **responsibility**.

How Can and Should We Attribute Moral Responsibility?

- When AI is used to make decisions for us and to do things for us, we encounter a problem that is shared with all automation technologies but which becomes even more important when AI enables us to delegate far more to machines than we used: **responsibility attribution**.
- If AI is given more agency and takes over what humans used to do, how do we then attribute moral responsibility?
- *If you have agency and if you can decide, you should take responsibility for your actions.*

How Can and Should We Attribute Moral Responsibility?

- Why not let the machine perform the actions and keep the responsibility on the side of the human?
  - An AI system may make its decisions and actions very quickly, e.g., in high- frequency trading or in a self-driving car, which gives the human too little time to make the final decision or to intervene.
  - AIs have histories: When the AI does things in a particular context of application, it may no longer be clear who created it, who used it first, and how the responsibility should be distributed among these different parties involved.
- **The solution**: to hold humans responsible for what the AI does, assuming that they know what the AI is doing and what they are doing with the AI.
- However, we do not always know exactly what the AI is doing and cannot always explain what it did or how it came to its decision.

Transparency and Explainability

- It is not transparent how the AI comes to its decision, and humans cannot fully explain the decision (e.g., the deep learning that uses neural networks).
- This is a problem for **responsibility**, since the humans who create or use the AI cannot explain a particular decision.
- While all automation technology raises problems of responsibility, here we encounter a problem specific to some kinds of AI: **the black box problem**.
- A lack of transparency leads to less trust in the technology and in the people who use the technology.
- Whether or not AI can directly provide those reasons and explanations, humans should be able to answer the question: "Why?"
- The challenge for AI researchers is to ensure that if an AI is used for decision making at all, the technology is built in such a way that humans will be able to answer that question.

What Needs to Be Done and Other Questions Policymakers Have to Answer

- what should be done, what precise course of action should be taken, why it should be done, when it should be done, how much should be done, by whom it should be done, and what the nature, extent, and urgency of the problem are.
- 1. it is important to justify the measures proposed.
- 2. one can try to make policy before the technology is fully developed and used.
- 3. fundamentally rethink society and its institutions, including the legal systems, in order to deal with the underlying problems and to prepare future generations
- 4. to distribute responsibility for policy and change: is it mainly up to governments to take action, or should businesses and industry develop their own course of action to ensure ethical AI?
- 5. how one defines the nature, extent, and urgency of the problem itself: a remarkable degree of convergence

Ethical Principles and Justifications

- The widely shared intuition that there is an urgency and importance in dealing with the ethical and societal challenges raised by AI has led to an avalanche of initiatives and policy documents.
- AI policy with an ethical component has been proposed by a range of actors,
  - governments and governmental bodies such as national ethics committees,
  - tech companies such as Google,
  - engineers and their professional organizations such as IEEE,
  - intergovernmental organizations such as the EU,
  - nongovernmental nonprofit actors, and researchers.

Ethical Principles and Justifications

- Many countries in Europe have AI strategies now that include an ethical component.
- "Explainable AI" is a goal shared by many policymakers.
  - The UK's House of Commons (2018) says that transparency and the right to explanation is key for algorithmic accountability, and that industries and regulators should tackle biased algorithmic decision making.
  - The European Group on Ethics in Science and New Technologies (EGE) releases a statement on AI, robotics, and autonomous systems: proposing the principles of human dignity, autonomy, responsibility, justice, equity, solidarity, democracy, rule of law and accountability, security and safety, data protection and privacy, and sustainability.

Ethical Principles and Justifications

- China has a national AI strategy.
  - China's development plan recognizes that AI is a disruptive technology that can affect social stability, impact law and social ethics, violate personal privacy, and create safety risks;
  - the plan therefore recommends to strengthen forward- looking prevention and minimizing risk (State Council of China 2017).
- In spite of cultural differences, it turns out that AI ethics policies are remarkably similar.
- Ideas such as ethics by design or value-sensitive design can help to create AI in a way that leads to more accountability, responsibility, and transparency.

Technological Solutions

- Some researchers have called for explainable artificial intelligence, for opening the black box.
- Explaining the rationale behind one's decision is not only ethically required but also an important aspect of human intelligence (Samek, Wiegand, and Müller 2017).
- The idea of explainable AI or transparent AI is then that the actions and decisions made by AIs should be easily understood.
- Ideas such as ethics by design or value-sensitive design can help to create AI in a way that leads to more accountability, responsibility, and transparency.
- Ethics by design could include the requirement that traceability is ensured at all stages (Dignum et al. 2018), thus contributing to the accountability of AI.